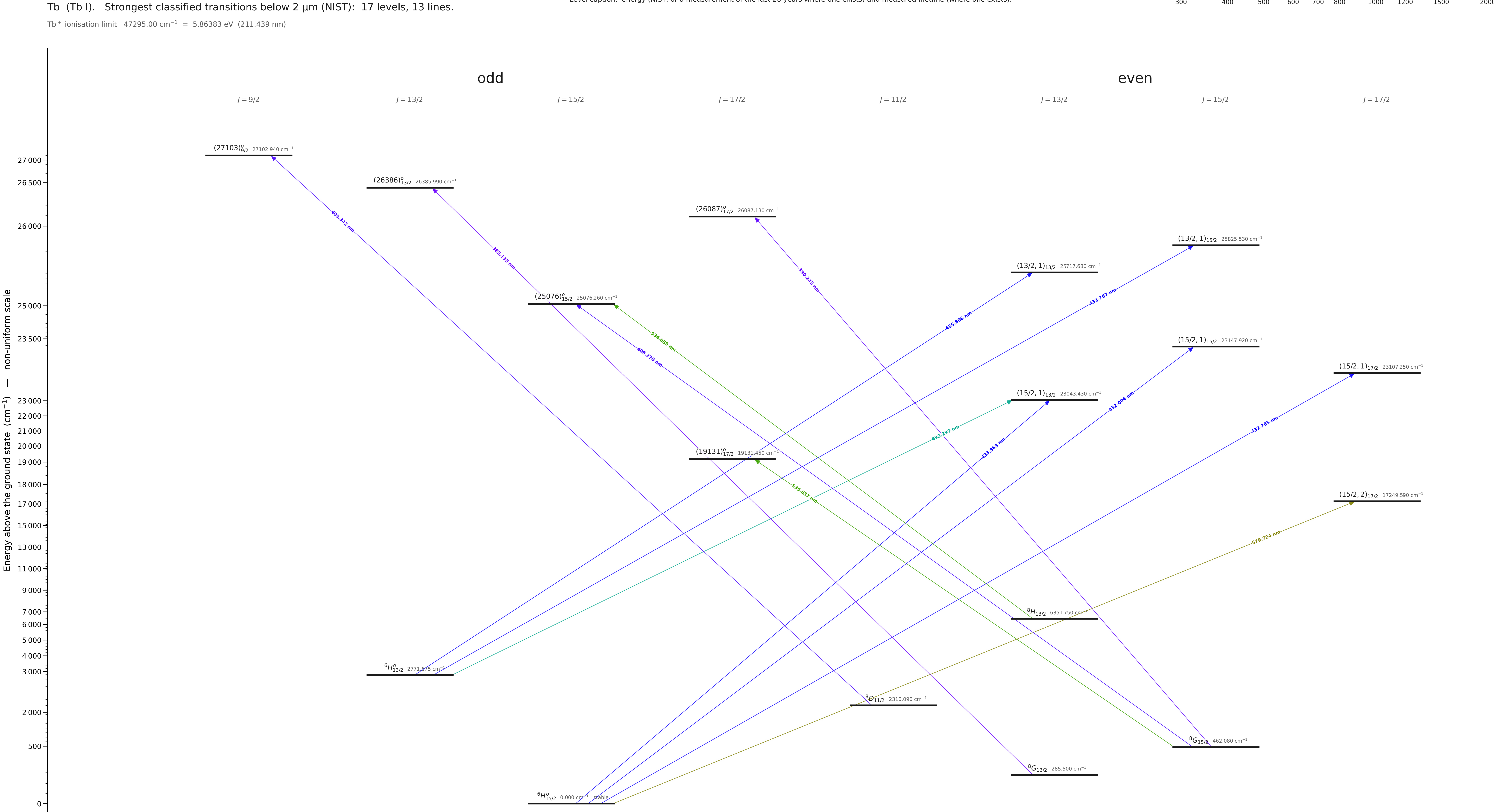
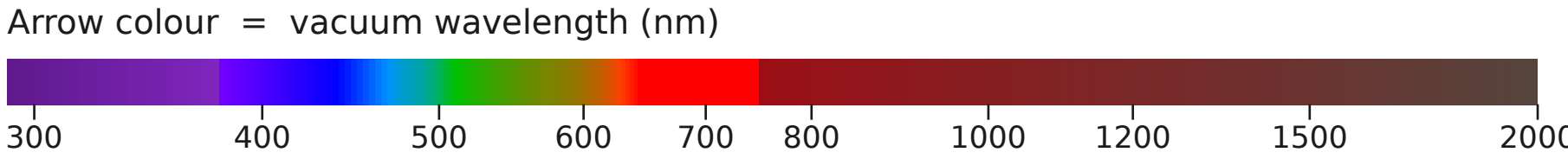


# Terbium

Arrow label: vacuum wavelength | reduced dipole matrix element  $| \langle J \| e r \| J' \rangle |$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                        | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $  \langle J \  e r \  J' \rangle  $ ( $e a_0$ ) | A ( $\text{s}^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|-----------------------------------|--------------------|--------------------|-------------|--------------------------------------------------|-----------------------|-------------------------|------------|--------------------------------------------------------------|
| $^6H_{15/2}^0 - (15/2, 2)_{17/2}$ | 579.7239           | 579.5632           | 517.1297    | -                                                | -                     | -                       | -          | first classified line between 433.8 and 1023.9 nm that co... |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## $^{159}\text{Tb}$ hyperfine structure ( $I = 3/2$ )

| Level               | A (MHz)      | B (MHz)      | Shift of each F level from the centre of gravity (MHz)     |
|---------------------|--------------|--------------|------------------------------------------------------------|
| $^6H_{15/2}^0$      | 673.7550(10) | 1449.445(12) | F=6: -8062.36 F=7: -4215.75 F=8: +1008.64 F=9: +7942.10    |
| $^8G_{13/2}$        | 532.2040(20) | 928.861(20)  | F=5: -5630.04 F=6: -3008.42 F=7: +591.97 F=8: +5421.20     |
| $^8G_{15/2}$        | 472.6430(20) | 1154.239(17) | F=6: -5605.73 F=7: -2989.77 F=8: +659.46 F=9: +5605.79     |
| $^8D_{11/2}$        | 405.106(3)   | -92.64(5)    | F=4: -3988.10 F=5: -1903.62 F=6: +542.17 F=7: +3318.97     |
| $^8H_{13/2}$        | 438.5(22)    | 1122(29)     | F=5: -4501.59 F=6: -2561.05 F=7: +357.41 F=8: +4555.88     |
| (15/2, 2) $_{17/2}$ | 857.1(4)     | 1758(5)      | F=7: -11599.67 F=8: -5776.98 F=9: +1762.41 F=10: +11367.52 |
| (15/2, 1) $_{13/2}$ | 685.7(20)    | 915(24)      | F=5: -7362.20 F=6: -3811.08 F=7: +865.65 F=8: +6914.32     |

Measured A, B constants; sources in the data file.